

Policy Gradient: REINFORCE, Variance Reduction, ~~Convergence~~

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CS 6789: Foundations of Reinforcement Learning

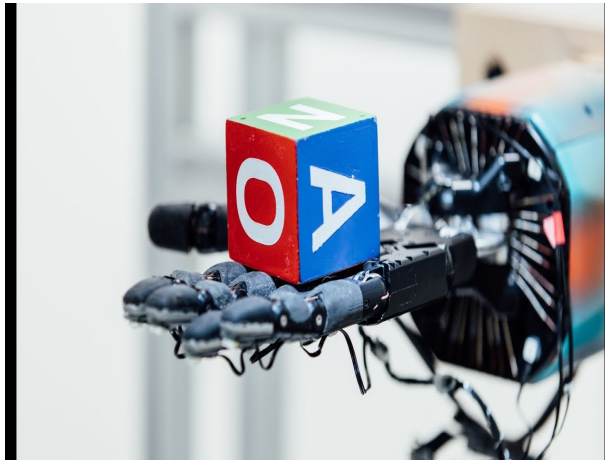
Policy Optimization



[AlphaZero, Silver et.al, 17]



[OpenAI Five, 18]



[OpenAI,19]

Recap: Infinite Horizon Discounted MDPs

1. Two formulations of Policy Gradient

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1. Two formulations of Policy Gradient

$$\nabla_{\theta} J(\pi_{\theta}) = \mathbb{E}_{\tau \sim \rho_{\theta}(\tau)} \left[\nabla_{\theta} \ln \rho_{\theta}(\tau) R(\tau) \right]$$

$$p_{\theta}(\tau) = \mu(s_0) \pi_{\theta}(a_0 | s_0) p(s_1 | a_0, s_0) \dots$$

Recap: Infinite Horizon Discounted MDPs

1. Two formulations of Policy Gradient

$$\nabla_{\theta} J(\pi_{\theta}) = \mathbb{E}_{\tau \sim \rho_{\theta}(\tau)} \left[\nabla_{\theta} \ln \rho_{\theta}(\tau) R(\tau) \right] = \mathbb{E}_{\tau \sim \rho_{\theta}(\tau)} \left[\left(\sum_{h=0}^{\infty} \nabla_{\theta} \ln \pi_{\theta}(a_h | s_h) \right) R(\tau) \right]$$

Recap: Infinite Horizon Discounted MDPs

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$$\nabla_{\theta} J(\pi_{\theta}) = \nabla_{\theta} \mathbb{E}_{s_0 \sim \rho} [V^{\pi_{\theta}}(s_0)]$$

Recap: Infinite Horizon Discounted MDPs

1. Two formulations of Policy Gradient

$$\nabla_{\theta} J(\pi_{\theta}) = \mathbb{E}_{\tau \sim \rho_{\theta}(\tau)} \left[\nabla_{\theta} \ln \rho_{\theta}(\tau) R(\tau) \right] = \mathbb{E}_{\tau \sim \rho_{\theta}(\tau)} \left[\left(\sum_{h=0}^{\infty} \nabla_{\theta} \ln \pi_{\theta}(a_h | s_h) \right) R(\tau) \right]$$

$$\nabla_{\theta} J(\pi_{\theta}) = \nabla_{\theta} \mathbb{E}_{s_0 \sim \rho} \left[V^{\pi_{\theta}}(s_0) \right] = \frac{1}{1 - \gamma} \mathbb{E}_{s, a \sim d^{\pi_{\theta}}} \nabla_{\theta} \ln \pi_{\theta}(a | s) Q^{\pi_{\theta}}(s, a)$$

Outline for today

1. Two formulations of Policy Gradient

2. Variance Reduction

Derivation of unbiased Stochastic Policy Gradient

$$\nabla_{\theta} J(\theta) := \frac{1}{1-\gamma} \mathbb{E}_{s,a \sim d^{\pi_{\theta}}} \left[\nabla_{\theta} \ln \pi_{\theta}(a | s) Q^{\pi_{\theta}}(s, a) \right]$$

$$d^{\pi}(s,a) = (1-\gamma) \sum_{h=0}^{\infty} \gamma^h p_h^{\pi}(s,a)$$

Derivation of unbiased Stochastic Policy Gradient

$$\nabla_{\theta} J(\theta) := \frac{1}{1-\gamma} \mathbb{E}_{s, a \sim d^{\pi_{\theta}}} \left[\nabla_{\theta} \ln \pi_{\theta}(a | s) Q^{\pi_{\theta}}(s, a) \right]$$

Draw $h \propto \gamma^h$, **roll-in** π_{θ} to generate $s_h, a_h \sim \mathbb{P}_h^{\pi_{\theta}}$

$$\mathbb{P}_0^{\pi} \dots \mathbb{P}_h^{\pi} \dots$$

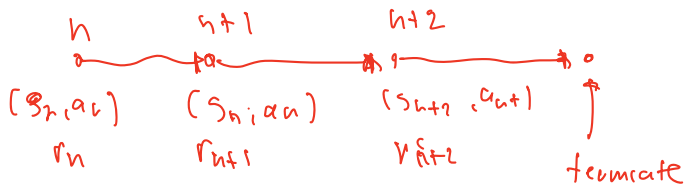
$\frac{\gamma}{1-\gamma} \quad \frac{\gamma^h}{1-\gamma}$

Derivation of unbiased Stochastic Policy Gradient

$$\nabla_{\theta} J(\theta) := \frac{1}{1-\gamma} \mathbb{E}_{s,a \sim d^{\pi_{\theta}}} \left[\nabla_{\theta} \ln \pi_{\theta}(a | s) \boxed{Q^{\pi_{\theta}}(s, a)} \right]$$

Draw $h \propto \gamma^h$, **roll-in** π_{θ} to generate $s_h, a_h \sim \mathbb{P}_h^{\pi_{\theta}}$

Roll-out π_{θ} from (s_h, a_h) : terminate with prob $1 - \gamma$, $\widetilde{Q}^{\pi_{\theta}}(s_h, a_h) = \sum_{\tau=h}^{t \geq h} r_{\tau}$



$$\mathbb{E}[\widetilde{Q}^{\pi_{\theta}}] = Q^{\pi_{\theta}}$$

$$\widetilde{Q}^{\pi_{\theta}}(s_h, a_h) = r_h + \gamma v_{h+1} + r_{h+2}$$

Derivation of unbiased Stochastic Policy Gradient

$$\nabla_{\theta} J(\theta) := \frac{1}{1-\gamma} \mathbb{E}_{s, a \sim d^{\pi_{\theta}}} \left[\nabla_{\theta} \ln \pi_{\theta}(a | s) Q^{\pi_{\theta}}(s, a) \right]$$

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Roll-out π_{θ} from (s_h, a_h) : terminate with prob $1 - \gamma$, $\widetilde{Q}^{\pi_{\theta}}(s_h, a_h) = \sum_{\tau=h}^{t \geq h} r_{\tau}$

Unbiased estimate: $\nabla_{\theta} \ln \pi_{\theta}(a_h | s_h) \widetilde{Q}^{\pi_{\theta}}(s_h, a_h)$

$$\mathbb{E} \left[\nabla_{\theta} \ln \pi_{\theta}(a_h | s_h) \widetilde{Q}^{\pi_{\theta}}(s_h, a_h) \right] = \nabla_{\theta} J(\theta)$$

Outline for today

1. Two formulations of Policy Gradient

2. Variance Reduction

~~3. Convergence of SGD~~

Variance Reduction via Action-Independent Baseline

Unbiased Estimate:

$$\nabla_{\theta} \ln \pi_{\theta}(a_h | s_h) \left(\widetilde{Q}^{\pi_{\theta}}(s_h, a_h) - \boxed{b(s_h)} \right)$$

Variance Reduction via Action-Independent Baseline

Unbiased Estimate:

$$\nabla_{\theta} \ln \pi_{\theta}(a_h | s_h) \left(\widetilde{Q}^{\pi_{\theta}}(s_h, a_h) - b(s_h) \right)$$

$$\mathbb{E}_{s, a \sim d^{\pi_{\theta}}} \left[\nabla_{\theta} \ln \pi_{\theta}(a | s) \right] b(s) = 0$$

Variance Reduction via Action-Independent Baseline

Unbiased Estimate:

$$\nabla_{\theta} \ln \pi_{\theta}(a_h | s_h) \left(\widetilde{Q}^{\pi_{\theta}}(s_h, a_h) - b(s_h) \right)$$

$$\mathbb{E}_{s, a \sim d^{\pi_{\theta}}} \left[\nabla_{\theta} \ln \pi_{\theta}(a | s) \right] b(s)$$

$$= \mathbb{E}_{s \sim d^{\pi_{\theta}}} \left[\boxed{b(s)} \boxed{\mathbb{E}_{a \sim \pi_{\theta}(\cdot | s)} \nabla_{\theta} \ln \pi_{\theta}(a | s)} \right]$$

Variance Reduction via Action-Independent Baseline

Unbiased Estimate:

$$\nabla_{\theta} \ln \pi_{\theta}(a_h | s_h) \left(\widetilde{Q}^{\pi_{\theta}}(s_h, a_h) - b(s_h) \right)$$

$$\begin{aligned} & \mathbb{E}_{s, a \sim d^{\pi_{\theta}}} \left[\nabla_{\theta} \ln \pi_{\theta}(a | s) \right] b(s) \\ &= \mathbb{E}_{s \sim d^{\pi_{\theta}}} \left[b(s) \underbrace{\mathbb{E}_{a \sim \pi_{\theta}(\cdot | s)} \left[\nabla_{\theta} \ln \pi_{\theta}(a | s) \right]}_{\frac{\nabla_{\theta} \pi_{\theta}(a | s)}{\pi_{\theta}(a | s)}} \right] = \mathbb{E}_{s \sim d^{\pi_{\theta}}} \left[b(s) \underbrace{\sum_a \pi_{\theta}(a | s)}_{\cancel{\pi_{\theta}(a | s)}} \frac{\nabla_{\theta} \pi_{\theta}(a | s)}{\cancel{\pi_{\theta}(a | s)}} \right] \end{aligned}$$

Variance Reduction via Action-Independent Baseline

Unbiased Estimate:

$$\nabla_{\theta} \ln \pi_{\theta}(a_h | s_h) \left(\widetilde{Q}^{\pi_{\theta}}(s_h, a_h) - b(s_h) \right)$$

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Variance Reduction via Action-Independent Baseline

Unbiased Estimate:

$$\nabla_{\theta} \ln \pi_{\theta}(a_h | s_h) \left(\widetilde{Q}^{\pi_{\theta}}(s_h, a_h) - b(s_h) \right)$$

$$\begin{aligned} & \mathbb{E}_{s,a \sim d^{\pi_\theta}} [\nabla_{\theta} \ln \pi_{\theta}(a | s)] b(s) \\ &= \mathbb{E}_{s \sim d^{\pi_\theta}} [b(s) \mathbb{E}_{a \sim \pi_{\theta}(\cdot | s)} \nabla_{\theta} \ln \pi_{\theta}(a | s)] = \mathbb{E}_{s \sim d^{\pi_\theta}} \left[b(s) \sum_a \pi_{\theta}(a | s) \frac{\nabla_{\theta} \pi_{\theta}(a | s)}{\pi_{\theta}(a | s)} \right] \\ &= \mathbb{E}_{s \sim d^{\pi_\theta}} \left[b(s) \sum_a \boxed{\nabla_{\theta} \pi_{\theta}(a | s)} \right] = \mathbb{E}_{s \sim d^{\pi_\theta}} \left[b(s) \boxed{\nabla_{\theta}} \left(\sum_a \pi_{\theta}(a | s) \right) \right] \\ &\quad \text{// } \underline{1} \\ &= \mathbb{E}_{s \sim d^{\pi_\theta}} [b(s) \nabla_{\theta} 1] = 0 \end{aligned}$$

Variance Reduction via Action-Independent Baseline

Unbiased Estimate:

$$\nabla_{\theta} \ln \pi_{\theta}(a_h | s_h) \left(\widetilde{Q}^{\pi_{\theta}}(s_h, a_h) - b(s_h) \right)$$

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Variance Reduction via Action-Independent Baseline

$$\nabla_{\theta} \ln \pi_{\theta}(a_h | s_h) \left(\widetilde{Q}^{\pi_{\theta}}(s_h, a_h) - b(s_h) \right)$$

The best baseline that minimizes variance:

$$\min_b \mathbb{E} \left[\left(\nabla_{\theta} \ln \pi_{\theta}(a_h | s_h) \left(\widetilde{Q}^{\pi_{\theta}}(s_h, a_h) - b(s_h) \right) \right)^{\top} \nabla_{\theta} \ln \pi_{\theta}(a_h | s_h) \left(\widetilde{Q}^{\pi_{\theta}}(s_h, a_h) - b(s_h) \right) \right]$$

Variance Reduction via Action-Independent Baseline

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$$b(s_h) = \frac{\mathbb{E} \left[\nabla_{\theta} \ln \pi_{\theta}(a_h | s_h)^{\top} \nabla_{\theta} \ln \pi_{\theta}(a_h | s_h) \widetilde{Q}^{\theta}(s_h, a_h) \right]}{\mathbb{E} \left[\nabla_{\theta} \ln \pi_{\theta}(a_h | s_h)^{\top} \nabla_{\theta} \ln \pi_{\theta}(a_h | s_h) \right]}$$

Variance Reduction via Action-Independent Baseline

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In practice:

$$b(s_h) = \boxed{V^{\pi_{\theta}}(s)}$$

Variance Reduction via Action-Independent Baseline

$$\nabla_{\theta} \ln \pi_{\theta}(a_h | s_h) \left(\widetilde{Q}^{\pi_{\theta}}(s_h, a_h) - b(s_h) \right)$$

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$$b(s_h) = \frac{\mathbb{E} \left[\nabla_{\theta} \ln \pi_{\theta}(a_h | s_h)^{\top} \nabla_{\theta} \ln \pi_{\theta}(a_h | s_h) \widetilde{Q}^{\theta}(s_h, a_h) \right]}{\mathbb{E} \left[\nabla_{\theta} \ln \pi_{\theta}(a_h | s_h)^{\top} \nabla_{\theta} \ln \pi_{\theta}(a_h | s_h) \right]}$$

In practice:

$$\nabla_{\theta} J(\theta) = \frac{1}{1-\gamma} \mathbb{E}_{s,a \sim d^{\pi_{\theta}}} \left[\nabla_{\theta} \ln \pi_{\theta}(a | s) (Q^{\pi_{\theta}}(s, a) - \boxed{V^{\pi_{\theta}}(s)}) \right] = \frac{1}{1-\gamma} \mathbb{E}_{s,a \sim d^{\pi_{\theta}}} \left(\nabla_{\theta} \ln \pi_{\theta}(a | s) A^{\pi_{\theta}}(s, a) \right)$$

$$Q^{\pi_{\theta}}(s, a) - V^{\pi_{\theta}}(s) = A^{\pi_{\theta}}(s, a)$$

Summary so far:

The most commonly used formulation:

Policy Gradient with V^{π_θ} as a baseline:

$$\nabla_\theta J(\theta) = \frac{1}{1-\gamma} \mathbb{E}_{s,a \sim d^{\pi_\theta}} \left[\nabla_\theta \ln \pi_\theta(a | s) A^{\pi_\theta}(s, a) \right]$$

Summary so far:

The most commonly used formulation:
Policy Gradient with V^{π_θ} as a baseline:

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Q: can you think about a way to get an unbiased estimate of $A^{\pi_\theta}(s, a)$ via one roll-out?

Summary

$$\nabla_{\theta} J(\theta) = \frac{1}{1-\gamma} \mathbb{E}_{s,a \sim d^{\pi_{\theta}}} \left[\nabla_{\theta} \ln \pi_{\theta}(a | s) (Q^{\pi_{\theta}}(s, a) - V_{\theta}^{\pi}(s)) \right]$$

Use an unbiased estimate of $\nabla_{\theta} J(\theta)$